

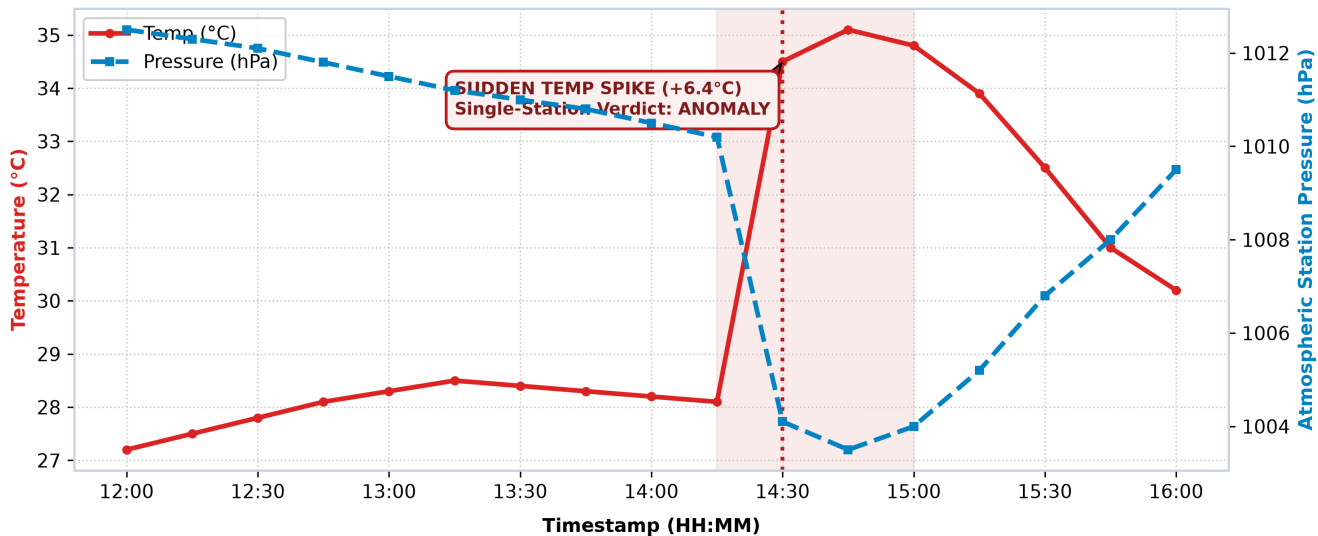
Scene 1: Isolated Single-Station Observation (The False Anomaly Trap)

Demonstrates how single-station threshold detectors trigger false disaster alerts during extreme local events

PRESENTER SCRIPT & VOICEOVER NARRATIVE:

"Look at the live stream on screen. Here is **Station 01** monitoring surface Ambient Temperature, Station Pressure, and Relative Humidity. At timestamp **T = 14:30**, notice how the temperature reading suddenly spikes by **+6.4°C** in a single observation interval! In isolation, traditional single-station algorithms look at this sharp excursion and immediately flag it as a sensor failure or hardware glitch. But is it really a fault?"

Station 01 (Isolated View) — Telemetry Ingestion Stream at Timestamp T = 14:30



Traditional Single-Station Detector Verdict: FAULT DETECTED (FALSE ALARM)

Why it fails: Single-station algorithms evaluate observations in total isolation. Without regional context, a sharp physical temperature jump is algorithmically indistinguishable from an electrical spike or sensor latch-up.

Physical Telemetry Parameters (Station 01)

- **Ambient Temperature (T):** 28.1°C → 34.5°C at T = 14:30
- **Station Pressure (P):** 1010.2 hPa → 1004.1 hPa (Barometric Dip)
- **Relative Humidity (RH):** 65.0% → 84.0% (Moisture Inflow)

Demonstrated System Limitation

Demonstrates why simple single-point rules trigger massive operator alert fatigue during dynamic weather events, proving the absolute necessity of spatial peer consensus.

Scene 2: Zoomed-Out Multi-Station Spatial Peer Consensus View

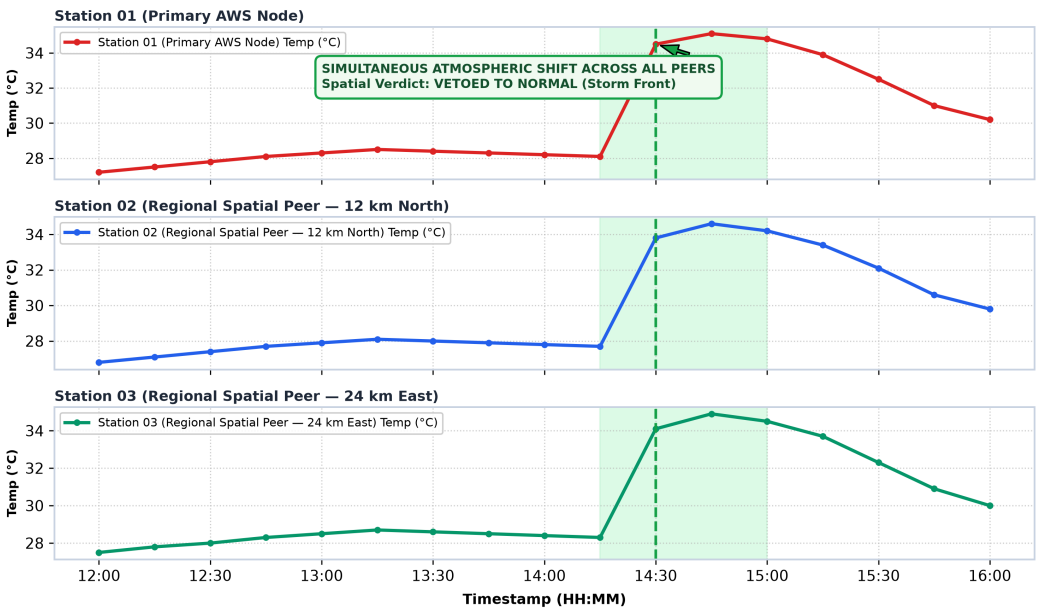
SkyGuard AI's Tier 5 Spatial Engine cross-checks regional peers to distinguish genuine weather fronts from faults



PRESENTER SCRIPT & VOICEOVER NARRATIVE:

"Now let's zoom out on screen to get broader spatial context! We bring in two neighboring stations from the same 50 km microclimate cluster: **Station 02** (12 km North) and **Station 03** (24 km East). Notice that at the exact same timestamp $T = 14:30$, both Station 02 and Station 03 record the exact same rapid thermodynamic shift! Because multiple independent physical sensors agree, this is NOT a hardware fault—it is a **genuine regional weather event**, such as an approaching storm front or gust outflow! **SkyGuard AI's Tier 5 Spatial Consensus Engine** automatically cross-checks regional peers and **VETOES the false alert back to NORMAL**."

Multi-Station Regional Spatial Peer Consensus — 50 km Microclimate Cluster Verification



✓ SkyGuard AI Tier 5 Spatial Consensus Verdict: VETOED TO NORMAL (GENUINE METEOROLOGICAL FRONT)

SkyGuard Resolution: Alert Vetoed | System Status: NORMAL | Spatial Consensus: 3 / 3 Cluster Stations Agree | Action: Telemetry passed clean to NWP forecast models without false maintenance dispatch.

Spatial Peer Cluster Verification (50 km Horizon)

- **Station 01 (Primary Node):** Temp spike +6.4°C (Confirmed)
- **Station 02 (12 km Peer):** Temp spike +6.1°C (Directionally Aligned)
- **Station 03 (24 km Peer):** Temp spike +5.8°C (Directionally Aligned)

Core Value Proposition for SIH Judges

SkyGuard AI does not just flag anomalies—it intelligently distinguishes between true sensor hardware failures and genuine extreme meteorological events, directly addressing the core requirement of the SIH AWS problem statement.